

### **Unit – 3 : Session – 3 : SLO – 2 | Synchronization and Inter-Thread Communication**

#### **Worksheet (Answers)**

1. Fill: The synchronized keyword is used to prevent RACE conditions between threads.
2. T/F: A synchronized block can only lock on this object. — FALSE (it can lock on any object reference)
3. MCQ: Method used for thread communication → A) notify()
4. Fill: The wait() method should be called inside a SYNCHRONIZED block or method.
5. Find the output: synchronized increment() called 2000 times total across two threads with join → A) 2000
6. T/F: notifyAll() wakes up all threads waiting on the object's monitor. — TRUE
7. MCQ: wait() throws an exception when → D) Both A and C (called outside synchronized context → IllegalMonitorStateException; thread interrupted → InterruptedException)
8. Fill: The notify() method wakes up one thread waiting on the OBJECT'S MONITOR.
9. Find the error: lock.wait() outside synchronized → C) IllegalMonitorStateException
10. Fill: The synchronized keyword can be applied to methods and BLOCKS.